| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 | Q8 |
|---|---|---|---|---|---|---|---|
| D | C | B | A | B | B | C | C |

12

Q9: $\frac{\sin^2 x}{2} - \frac{5\sin^4 x}{12}$

Q10: [illegible]

Q11: $(\ln x \cdot \frac{x^2}{2}) - \frac{x^2}{4}$

3

12) $\int \frac{dx}{x^2+2x-15} \rightarrow \int \frac{dx}{(x^2+2x+1)-1-15} \rightarrow \frac{-1}{16}\int \frac{1}{(x+1)(x+1)}$ — —

$\frac{-1}{16}\int \frac{a}{(x+1)} \cdot \frac{b}{(x+1)}\, dx$

$\frac{-1}{16}\left(a\ln(x+1) \cdot b\ln(x+1)\right)$

$\frac{-1}{16}\left(\frac{1}{6}\ln(x+1) \cdot (0\ln(x+1))\right)$

$1 = a(x+1) + b(x+1)$

$1 = (a+b)(x+1)$

$\frac{1}{x+1} = a+b$

$a+b = \frac{1}{2}$

$a = \frac{1}{2} - b$

$b = \frac{1}{6} - a$

$a = \frac{3}{6} - \frac{1}{6} - a$

$a+a = \frac{2}{6} \rightarrow 2a = \frac{2}{6}$

$a = \frac{1}{6}$

$b = \frac{1}{6} - \frac{1}{6} = 0$

Best wishes /



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Part A: Multiple choice (16 / marks)

*-Be sure that you have 8 different questions.*
*-Complete the following table by putting the correct answer letter in the table.*

Q1) $\lim_{x\to 0}\left(1+\frac{x}{4}\right)^{\frac{8}{x}}$ = a) $e$ b) $e^{-7}$ c) $e^{6}$ d) $e^{2}$

Q2) the range of the function f(x) = tanh x is
a) (1,2) b) (-1,3) c)(-1,1) d)(0,2)

Q3) $\int \sinh^5 x \cosh x\,dx$ a) $\frac{\cosh^6 x}{6}$ b) $\frac{\sinh^6 x}{6}+c$ c) $\frac{\sinh x}{6}$ d) $\sinh^6 x$

Q4) $\int_{-\sqrt{3}}^{\sqrt{3}} \sqrt{3-x^2}\,dx$ a) $\frac{3\pi}{2}$ b) $3\pi$ c) $\frac{2\pi}{3}$ d) $\sqrt{3}\pi$

Q5) Find the Arc length of the given curve x=2 cos t, y=2sin t, $0 \le t \le \pi/2$

a) $\pi$ b) $\pi/2$ c) $3\pi$ d) $4\pi$

Q6) Find the volume of the solid generated when the region enclosed between the curves $y=\sqrt{25-x^2}$, $y=\sqrt{24}$ revolved about the x-axis

a) $\pi$ b) $4\pi/3$ c) $3\pi/5$ d) $4\pi$

Q7) $\lim_{x\to 0^+} x\ln x$ = a) 1 b) 2 c)0 d)-1

Q8) $\lim_{x\to\infty}\left(1-\frac{3}{x}\right)^{2x}$ = a) $e$ b) $e^{-7}$ c) $e^{6}$ d) $e^{-6}$

Part B: Fill the Blanks (9 - marks)

Q9: $\int \sin^2 x \cos^3 x\,dx=$

Q10: $\int \sin 3x \sin 2x\,dx=$

Q11: $\int x\ln x\,dx=$

Part C: Written Questions (5 / marks)

Question 12: Find $\int \frac{dx}{x^2+2x-15}$

Best Wishes /



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| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 | Q8 |
|---|---|---|---|---|---|---|---|
| D | C | B | A | A | B | C | D |

Q9: $\frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} + C$

Q10: $\frac{1}{2}\left[\sin x - \frac{\sin 5x}{5}\right] + C$

Q11: $\frac{1}{2} x^2 \ln x - \frac{x^2}{4} + C$

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12)

$$\int \frac{dx}{x^2+2x-15} = \int \frac{dx}{(x-3)(x+5)} = \int \frac{A}{(x-3)} dx + \int \frac{B}{x}$$

$$A \ln|x-3| + B \ln|x+5| + C$$

To Find A, B

$$A(x+5) + B(x-3) = 1$$

let $x=3$

$8A = 1$

$A = \frac{1}{8}$

let $x=-5$

$-8B = 1$

$B = \frac{-1}{8}$



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